

# FMS - Product Requirements Document

A comprehensive Fleet Management System (FMS) for logistics and transportation operations.

## 1. Executive Summary

This PRD documents the requirements for a web-based Fleet Management System (FMS). The application is a comprehensive logistics and transportation management platform that enables organizations to manage their fleet operations, track shipments, handle customer and vendor relationships, manage inventory, and process financial transactions.

## 2. Screen Inventory & Summaries

### 2.1 Dashboard & Overview Screens

| Screen    | Description                                                                                                                               |
|-----------|-------------------------------------------------------------------------------------------------------------------------------------------|
| Dashboard | Main landing page with summary statistics, key metrics, and quick navigation. Shows shipment counts, pending orders, and recent activity. |
| Home      | Alternative entry point with navigation tiles to main modules.                                                                            |

### 2.2 Shipment Management Screens

| Screen         | Description                                                                                                                                                                                                       |
|----------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Shipments List | Data grid displaying all shipments with columns: Shipment ID, Customer, Origin, Destination, Status, Date, Weight, Volume, Carrier, Tracking Number. Includes search, filter, sort, pagination, and bulk actions. |

|                                   |                                                                                                                                                                                                                                                                                              |
|-----------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Shipment Details Form</b>      | Form for creating/editing shipments with fields: Shipment ID, Customer Name, Vendor Name, Origin, Destination, Shipment Date, Expected Delivery Date, Package Type, Weight, Dimensions, Quantity, Description, Current Status, Tracking Number. Tabbed interface with "Details" tab visible. |
| <b>Forwarding Job Form</b>        | Comprehensive freight forwarding form with party details (Shipper, Consignee, Agents), routing (POL, POD, Carrier, Vessel, Voyage), cargo specifications (weights, dimensions, marks), and transport mode selection. Multi-section collapsible layout.                                       |
| <b>Shipment Events/Milestones</b> | Timeline view for tracking shipment progress with milestone events (Booked, Received, Departed, Arrived, Delivered), planned vs actual dates, status indicators, and event notes.                                                                                                            |
| <b>Shipment Tracking</b>          | Grid view with enhanced tracking fields: Reference Number, Pickup Date, Delivery Date, Equipment Type, Sales Rep, Charges.                                                                                                                                                                   |

## 2.3 Customer Management Screens

| Screen                  | Description                                                                                                                     |
|-------------------------|---------------------------------------------------------------------------------------------------------------------------------|
| <b>Customers List</b>   | Grid showing customer records with columns: ID, Customer Name, Contact Person, Phone Number, Status. CRUD operations available. |
| <b>Customer Details</b> | Form with customer information, address fields, and contact details.                                                            |

## 2.4 Vendor Management Screens

| Screen              | Description                                                         |
|---------------------|---------------------------------------------------------------------|
| <b>Vendors List</b> | Grid displaying vendor/carrier information for shipment assignment. |

## 2.5 Order Management Screens

|  |  |
|--|--|
|  |  |
|--|--|

| Screen               | Description                                                                                              |
|----------------------|----------------------------------------------------------------------------------------------------------|
| <b>Orders List</b>   | Grid with columns: Order ID, Customer Name, Order Date, Due Date, Status, Amount, Priority, Assigned To. |
| <b>Order Details</b> | Form for order creation with Product, Quantity, and status fields.                                       |

## 2.6 Inventory Screens

| Screen                | Description                                                                                         |
|-----------------------|-----------------------------------------------------------------------------------------------------|
| <b>Inventory List</b> | Grid showing inventory items with columns: ID, Name, Description, Status, Type, Category, Quantity. |

## 2.7 Financial Screens

| Screen                 | Description                                                                                                                                                                           |
|------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Invoicing</b>       | AR module with invoice listing: Invoice Number, Customer, Date, Due Date, Amount, Currency, Outstanding Balance, Status, Age (Days). Supports multi-currency and job cost allocation. |
| <b>Invoice Details</b> | Line item entry with charge codes, quantities, unit prices, tax calculations, and job reference allocation.                                                                           |
| <b>Collections</b>     | Payment tracking with aging reports (Current, 1-30, 31-60, 60+ days), payment matching, and reminder workflows.                                                                       |
| <b>Aging Reports</b>   | Summary of outstanding receivables by customer with age buckets and totals.                                                                                                           |

## 2.8 Reports & Analytics

| Screen                   | Description                                                        |
|--------------------------|--------------------------------------------------------------------|
| <b>Reports Dashboard</b> | Analytics section with summary panels, charts, and data exports.   |
| <b>Custom Reports</b>    | Configurable report generation with date range and filter options. |

## 2.9 Administrative Screens

| Screen          | Description                                      |
|-----------------|--------------------------------------------------|
| Settings        | Application configuration and preferences.       |
| User Management | Admin section for user accounts and permissions. |
| Security        | Access control and authentication settings.      |

## 3. Core Data Entities & Schema

### 3.1 Shipment Schema

```
interface Shipment {
  // Primary Fields
  id: string; // Unique identifier (e.g., "SHP-001")
  referenceNumber: string; // External reference number
  trackingNumber: string; // Carrier tracking number

  // Relationships
  customerId: string; // FK to Customer
  vendorId?: string; // FK to Vendor/Carrier
  salesRepId?: string; // FK to User (sales representative)
  assignedToId?: string; // FK to User

  // Location Information
  origin: string; // Origin city/location
  originAddress?: Address; // Detailed origin address
  destination: string; // Destination city/location
  destinationAddress?: Address; // Detailed destination address

  // Dates
  shipmentDate: Date; // Date shipment created
  pickupDate?: Date; // Scheduled pickup date
  expectedDeliveryDate?: Date; // Estimated delivery date
  actualDeliveryDate?: Date; // Actual delivery date
  dueDate?: Date; // Due date for delivery
  createdAt: Date; // Record creation timestamp
  updatedAt: Date; // Last modification timestamp

  // Package Information
  packageType?: string; // Type of packaging
  weight: number; // Weight in kg/lbs
  volume: number; // Volume in m³/ft³
  dimensions?: Dimensions; // L x W x H
```

```

quantity: number; // Number of pieces/packages
description?: string; // Package description

// Service & Pricing
serviceType: string; // Service level (express, standard, etc.)
equipmentType?: string; // Container type, vehicle type
charges: number; // Total charges/cost
currency: string; // Currency code (USD, EUR, etc.)

// Status & Priority
status: ShipmentStatus; // Current status
priority: Priority; // Priority level

// Maritime-specific (optional)
containerNumber?: string; // Container ID
vesselName?: string; // Ship name
voyageNumber?: string; // Voyage reference
}

type ShipmentStatus =
  | "Draft"
  | "Pending"
  | "Booked"
  | "In Transit"
  | "Delivered"
  | "Cancelled"
  | "On Hold";

type Priority = "Low" | "Medium" | "High" | "Urgent";

interface Dimensions {
  length: number;
  width: number;
  height: number;
  unit: "cm" | "in" | "m";
}

interface Address {
  street: string;
  city: string;
  state: string;
  country: string;
  postalCode: string;
}

```

## 3.2 Customer Schema

```

interface Customer {
  id: string; // Unique identifier

```

```

name: string; // Company/customer name
code?: string; // Customer code

// Contact Information
contactPerson: string; // Primary contact name
email: string; // Email address
phoneNumber: string; // Phone number

// Address
address: Address; // Primary address
billingAddress?: Address; // Billing address

// Classification
type?: string; // Customer type/segment
category?: string; // Business category

// Status & Audit
status: "Active" | "Inactive" | "Suspended";
createdAt: Date;
updatedAt: Date;
createdBy?: string; // FK to User
}

```

### 3.3 Vendor/Carrier Schema

```

interface Vendor {
  id: string;
  name: string;
  code?: string;

  // Contact Information
  contactPerson: string;
  email: string;
  phoneNumber: string;
  address: Address;

  // Classification
  type: string; // e.g., 'Carrier', 'Supplier'
  serviceTypes: string[]; // Services offered

  // Status & Audit
  status: "Active" | "Inactive";
  createdAt: Date;
  updatedAt: Date;
}

```

### 3.4 Order Schema

```
interface Order {
  id: string; // Order ID
  referenceNumber?: string; // External reference

  // Relationships
  customerId: string; // FK to Customer
  shipmentId?: string; // FK to Shipment (when shipped)
  assignedToId?: string; // FK to User

  // Order Details
  orderDate: Date; // Order creation date
  dueDate?: Date; // Expected completion date

  // Line Items
  items: OrderItem[]; // Products/services ordered

  // Financial
  amount: number; // Total order amount
  currency: string; // Currency code

  // Status
  status: OrderStatus;
  priority: Priority;

  // Audit
  createdAt: Date;
  updatedAt: Date;
  createdBy?: string;
}
```

```
interface OrderItem {
  id: string;
  productId?: string; // FK to Product/Inventory
  productName: string;
  description?: string;
  quantity: number;
  unitPrice: number;
  total: number;
}
```

```
type OrderStatus =
  | "Draft"
  | "Pending"
  | "Confirmed"
  | "Processing"
  | "Shipped"
  | "Completed"
  | "Cancelled";
```

## 3.5 Inventory Schema

```
interface InventoryItem {
  id: string;
  name: string;
  description?: string;
  code?: string; // SKU or product code

  // Classification
  type: string; // Product type
  category: string; // Product category

  // Quantities
  quantity: number; // Current stock quantity
  unitOfMeasure: string; // e.g., 'pcs', 'kg', 'boxes'

  // Pricing
  unitPrice?: number;

  // Location
  location?: string; // Warehouse location

  // Status & Audit
  status: "Active" | "Inactive" | "Discontinued";
  createdAt: Date;
  updatedAt: Date;
}
```

## 3.6 User Schema

```
interface User {
  id: string;
  username: string;
  email: string;
  passwordHash: string;

  // Profile
  firstName: string;
  lastName: string;
  avatarUrl?: string;
  initials?: string; // e.g., "JH", "MS"

  // Authorization
  role: UserRole;
  permissions: string[];

  // Status & Audit
  status: "Active" | "Inactive" | "Suspended";
}
```



```
    lastLogin?: Date;  
    createdAt: Date;  
    updatedAt: Date;  
}
```

```
type UserRole =  
  | "Super Admin"  
  | "Admin"  
  | "Manager"  
  | "Sales Rep"  
  | "Operator"  
  | "Viewer";
```

### 3.7 Invoice Schema (Enhanced)

```
interface Invoice {  
  id: string;  
  invoiceNumber: string;  
  
  // Relationships  
  customerId: string;  
  jobId?: string; // FK to ForwardingJob  
  orderId?: string;  
  shipmentId?: string;  
  
  // Dates  
  invoiceDate: Date;  
  dueDate: Date;  
  paymentDate?: Date; // When fully paid  
  
  // Financial  
  currency: string; // ISO currency code  
  exchangeRate: number; // Rate to local currency  
  subtotal: number; // Before tax  
  taxAmount: number;  
  totalAmount: number; // Total invoice amount  
  amountPaid: number; // Amount received  
  outstandingBalance: number; // Remaining balance  
  
  // Terms  
  paymentTerms: string; // e.g., "Net 30", "COD"  
  
  // Status & Aging  
  status: InvoiceStatus;  
  ageDays: number; // Days since invoice date  
  
  // Line Items  
  lineItems: InvoiceLineItem[];
```

```

    // Audit
    createdAt: Date;
    updatedAt: Date;
    createdBy: string;
}

type InvoiceStatus =
    | "Draft"
    | "Open"
    | "Sent"
    | "Partial" // Partially paid
    | "Paid"
    | "Overdue"
    | "Cancelled"
    | "Disputed";

interface InvoiceLineItem {
    id: string;
    invoiceId: string;

    chargeCode: string; // Revenue/charge type code
    description: string;
    quantity: number;
    unitPrice: number;
    totalAmount: number;
    taxCode?: string;
    taxAmount?: number;

    // Job allocation
    jobId?: string;
    costAllocation?: string;
}

interface AgingReport {
    customerId: string;
    customerName: string;
    totalOutstanding: number;
    current: number; // Not yet due
    days1to30: number;
    days31to60: number;
    days61to90: number;
    days91Plus: number;
}

```

### 3.8 Forwarding Job Schema

```

interface ForwardingJob {
    id: string;
    jobNumber: string; // Unique job reference
}

```

```
referenceNumbers: string[]; // Multiple external references

// Transport
transportMode: "Sea" | "Air" | "Road" | "Rail" | "Multimodal";
serviceType: string; // FCL, LCL, Express, etc.
direction: "Import" | "Export" | "Crossship" | "Domestic";

// Parties (FK references)
shipperId: string; // FK to Customer (shipper)
consigneeId: string; // FK to Customer (consignee)
notifyPartyId?: string; // FK to Customer
originAgentId?: string; // FK to Vendor/Agent
destinationAgentId?: string; // FK to Vendor/Agent
carrierId?: string; // FK to Vendor (carrier)

// Routing
placeOfReceipt?: string; // Initial pickup location
portOfLoading: string; // POL
portOfDischarge: string; // POD
finalDestination?: string; // Final delivery location

// Vessel/Flight Details
vesselName?: string;
voyageNumber?: string;
flightNumber?: string;

// Dates
etd?: Date; // Estimated Time of Departure
eta?: Date; // Estimated Time of Arrival
atd?: Date; // Actual Time of Departure
ata?: Date; // Actual Time of Arrival

// Cargo
goodsDescription: string;
packageCount: number;
packageType: string; // Cartons, Pallets, etc.
grossWeight: number;
netWeight?: number;
volume: number; // CBM
marksAndNumbers?: string;

// Container (if applicable)
containerNumber?: string;
sealNumber?: string;
containerType?: string; // 20GP, 40HC, etc.

// Status
status: JobStatus;

// Audit
createdAt: Date;
```

```

    updatedAt: Date;
    createdBy: string;
}

```

```

type JobStatus =
  | "Draft"
  | "Confirmed"
  | "In Progress"
  | "At Origin"
  | "In Transit"
  | "At Destination"
  | "Delivered"
  | "Closed"
  | "Cancelled";

```

### 3.9 Shipment Event Schema

```

interface ShipmentEvent {
  id: string;
  shipmentId: string; // FK to Shipment/ForwardingJob

  eventCode: MilestoneCode; // Standard event code
  eventDescription: string; // Human-readable description
  eventType: "Milestone" | "Document" | "Financial" | "Exception" | "Customs";

  plannedDateTime?: Date; // Expected date/time
  actualDateTime?: Date; // Actual date/time (when completed)

  location?: string; // Where event occurred
  status: EventStatus;
  isRequired: boolean; // Must this milestone be completed?

  notes?: string; // Additional comments
  attachmentUrl?: string; // Related document

  // Audit
  createdBy: string;
  createdAt: Date;
  updatedBy?: string;
  updatedAt?: Date;
}

type MilestoneCode =
  | "BKD" // Booked
  | "RCV" // Cargo Received
  | "GIN" // Gate In
  | "DOC" // Documents Received
  | "CUS" // Customs Submitted
  | "CLR" // Customs Cleared

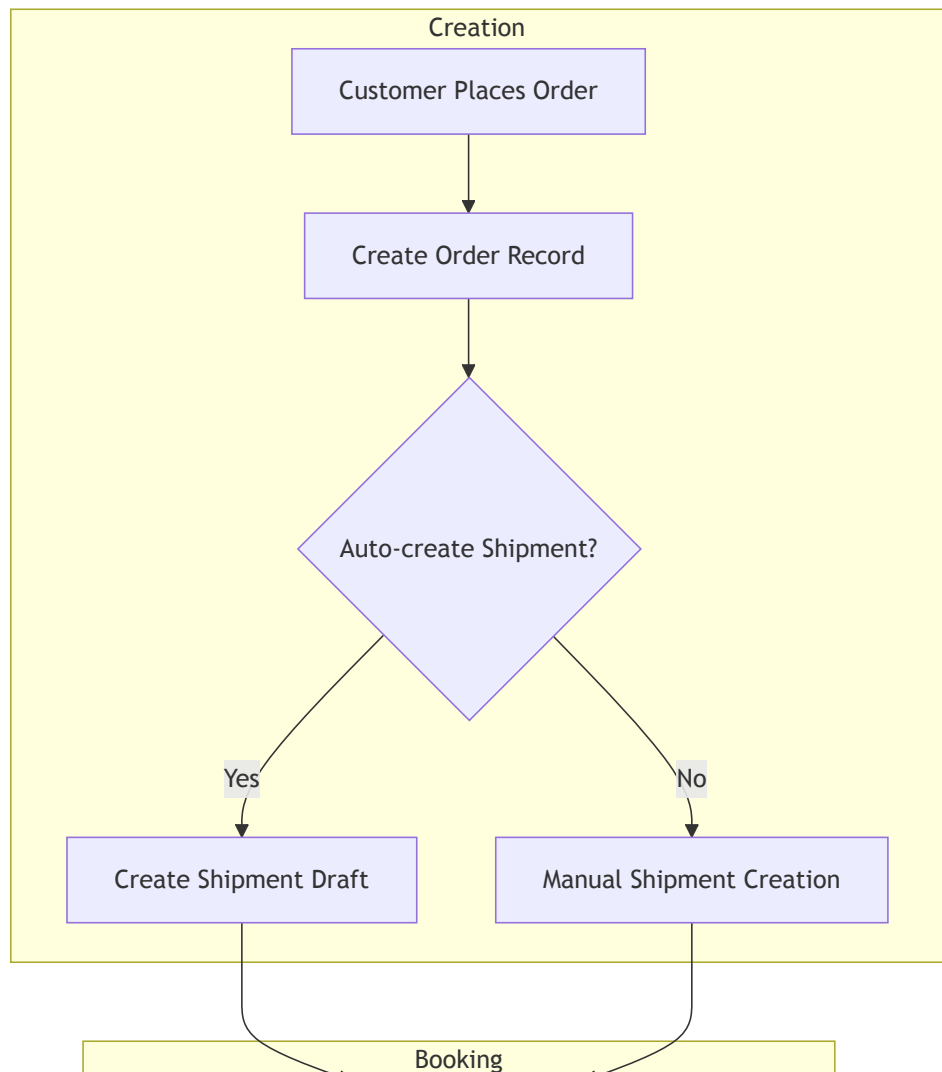
```

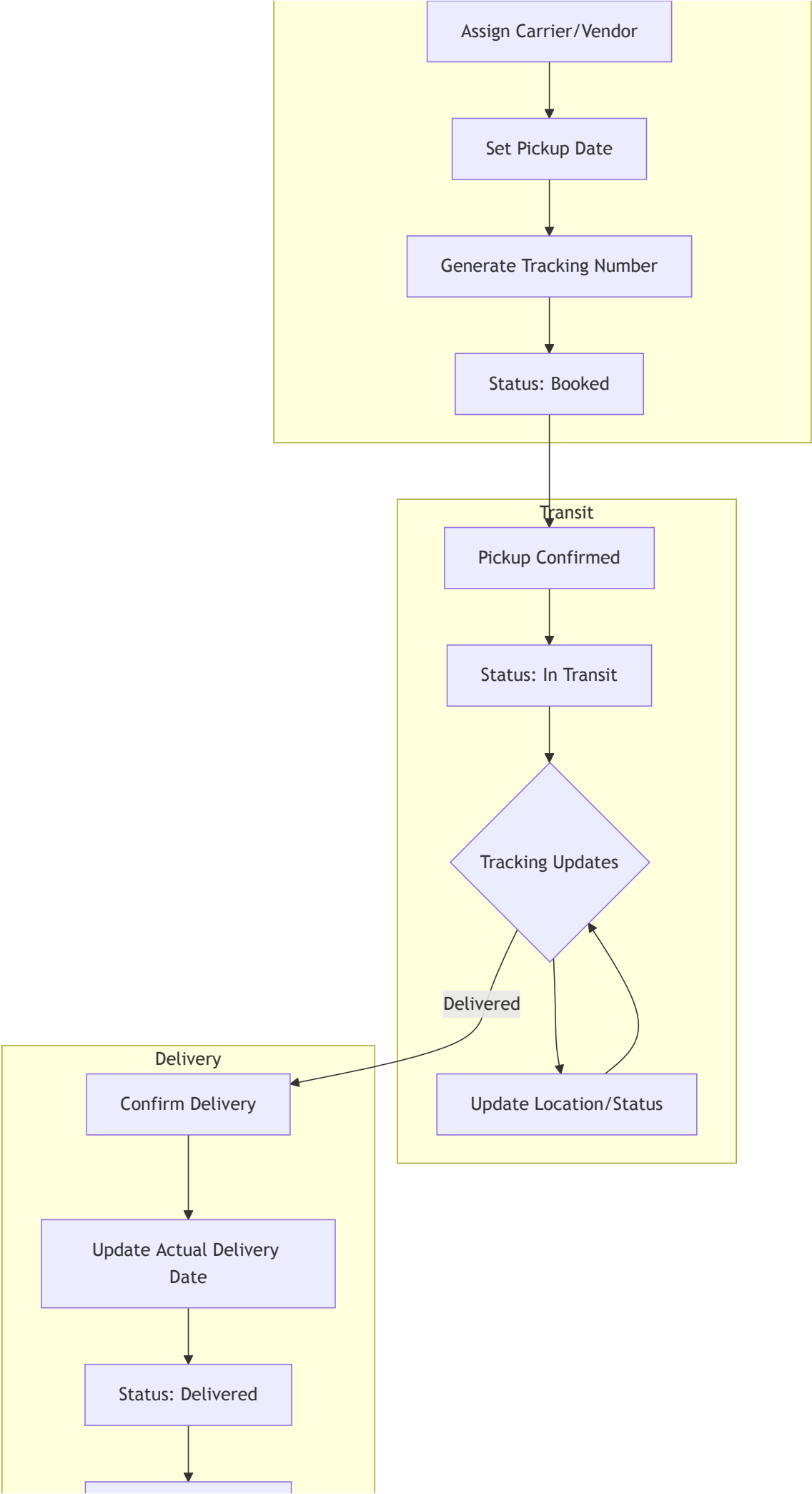
```
| "LOD" // Loaded  
| "DEP" // Departed  
| "ARR" // Arrived  
| "DIS" // Discharged  
| "AVL" // Available for Pickup  
| "OFD" // Out for Delivery  
| "DLV" // Delivered  
| "POD"; // Proof of Delivery Received
```

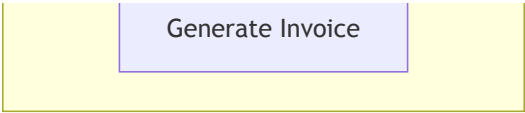
```
type EventStatus =  
| "Pending"  
| "Completed"  
| "Overdue"  
| "Skipped"  
| "Cancelled";
```

## 4. Data Flow Diagrams

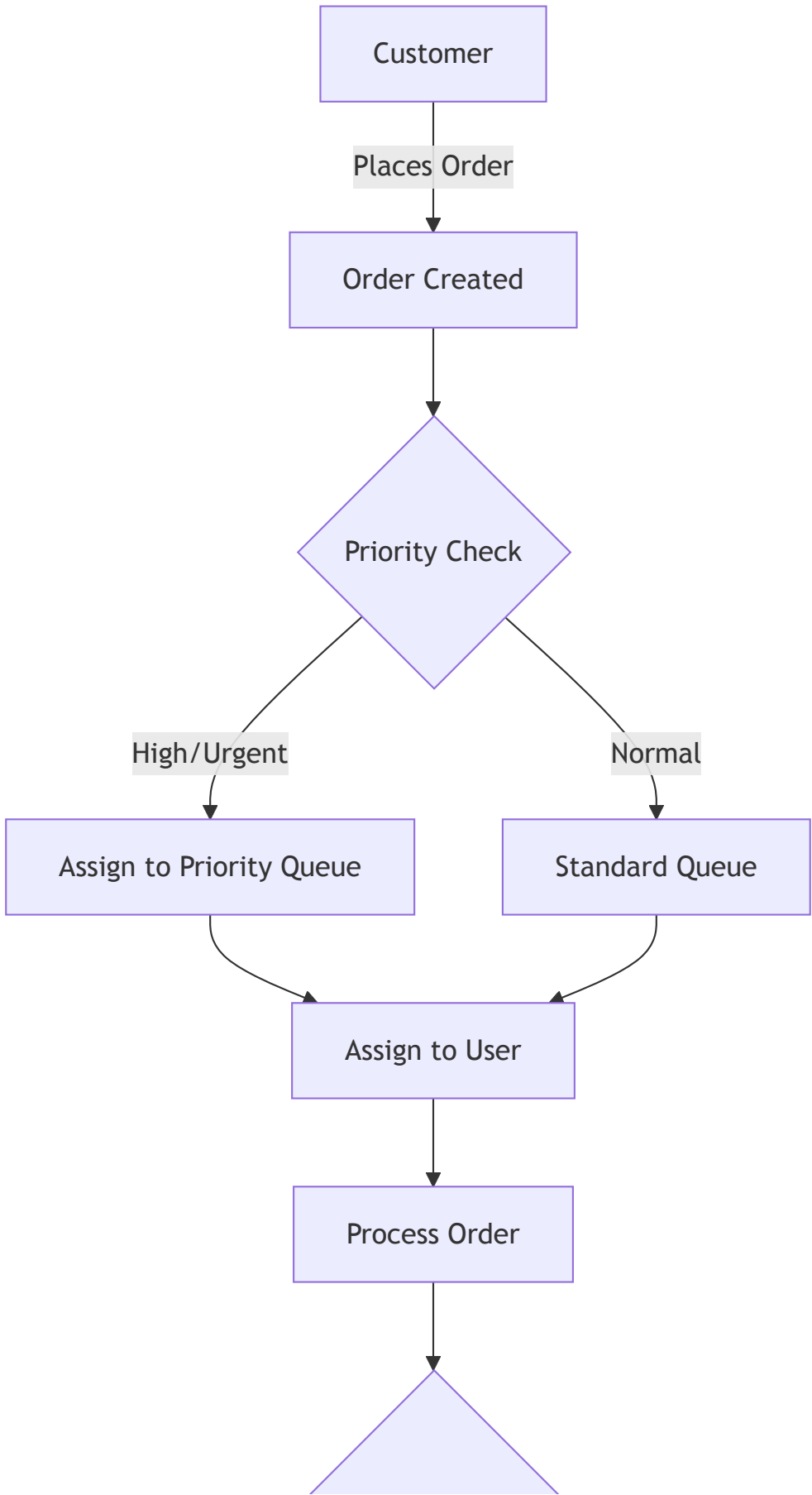
### 4.1 Shipment Lifecycle Flow

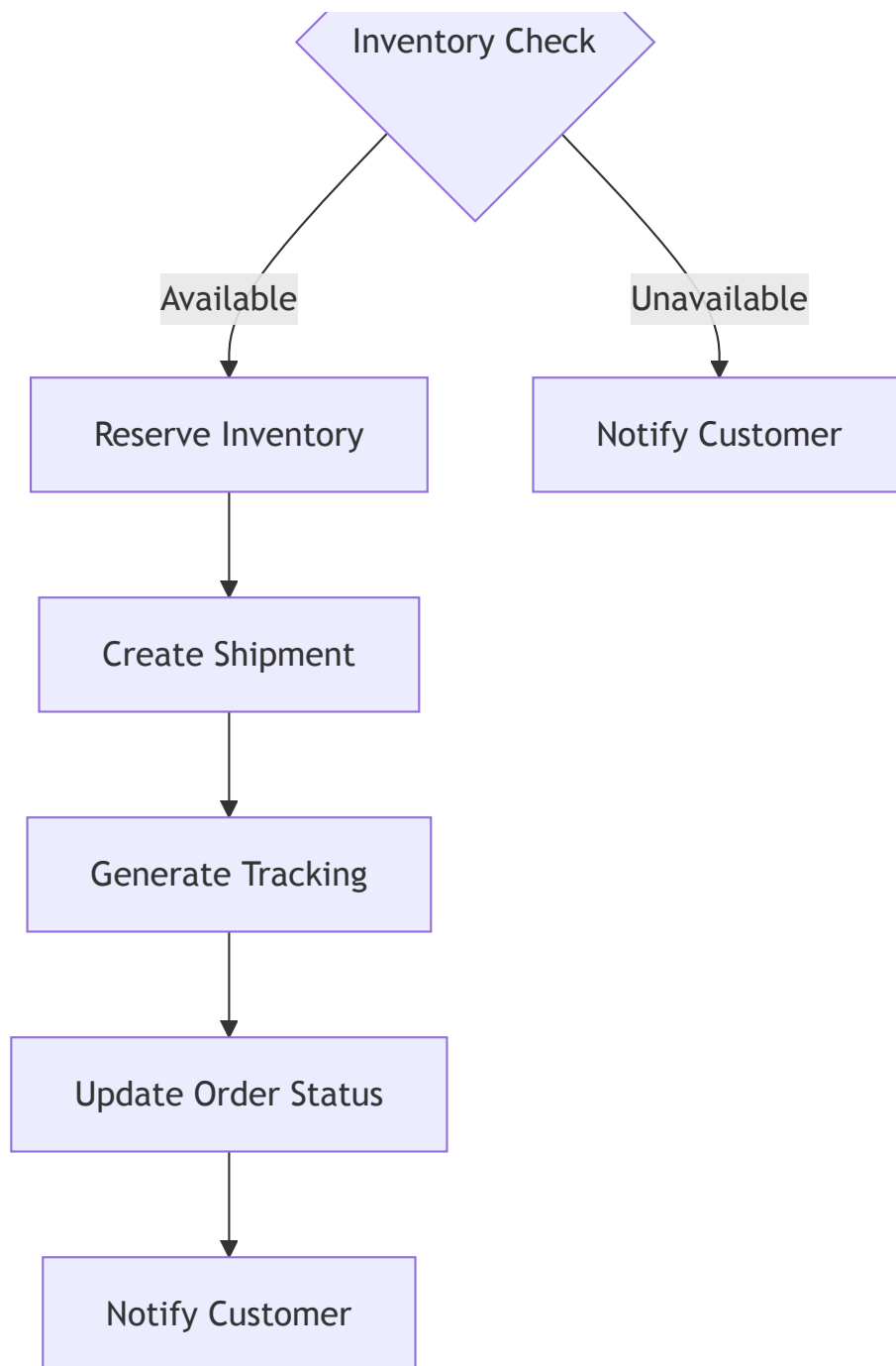




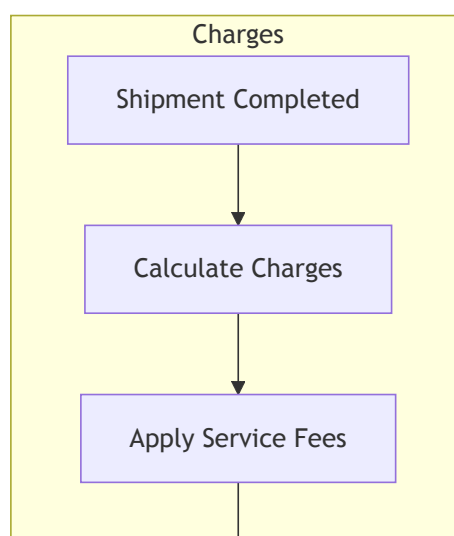


## 4.2 Order Processing Flow

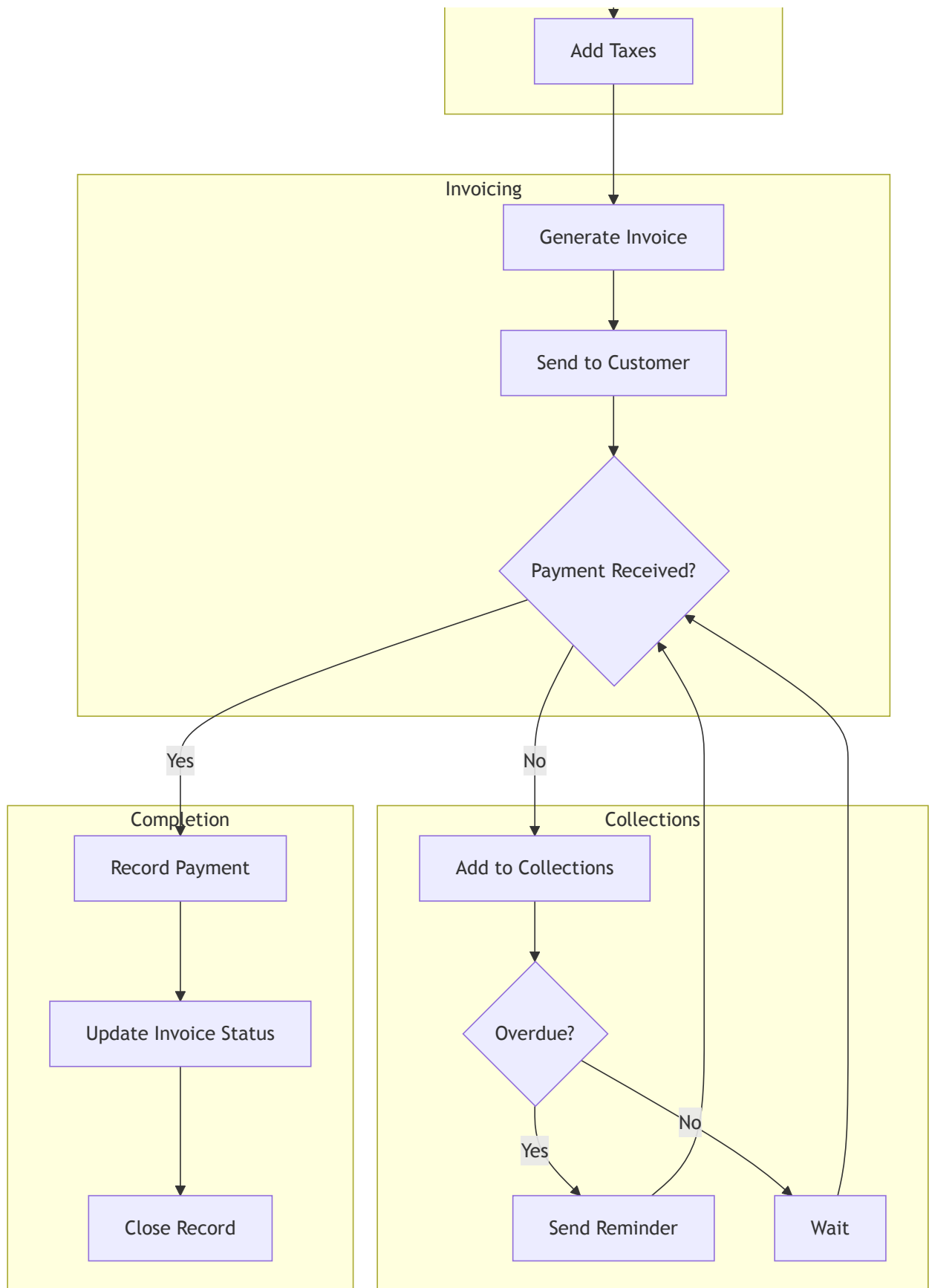




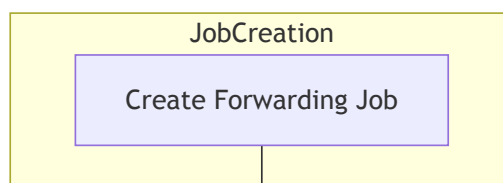
## 4.3 Financial Flow

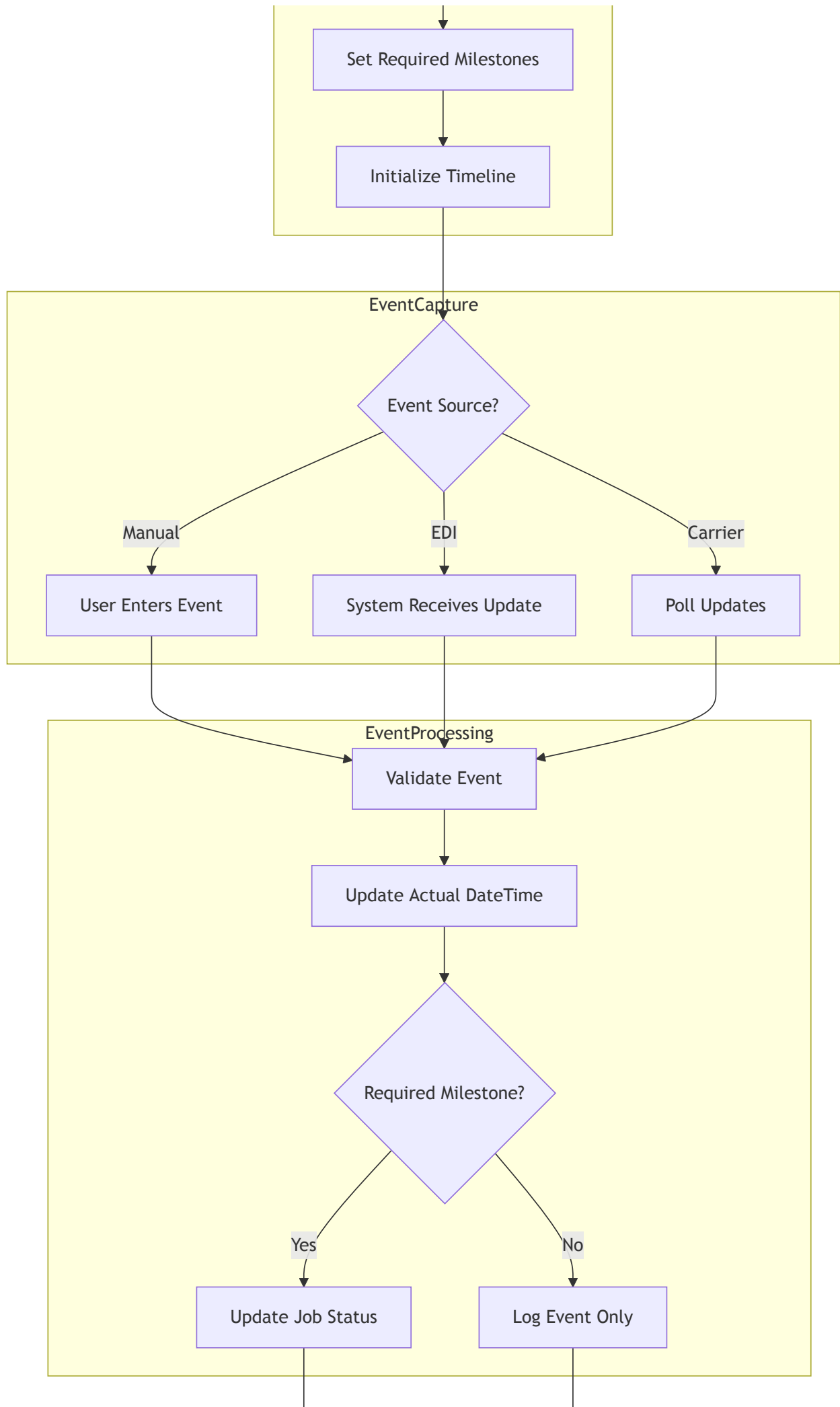


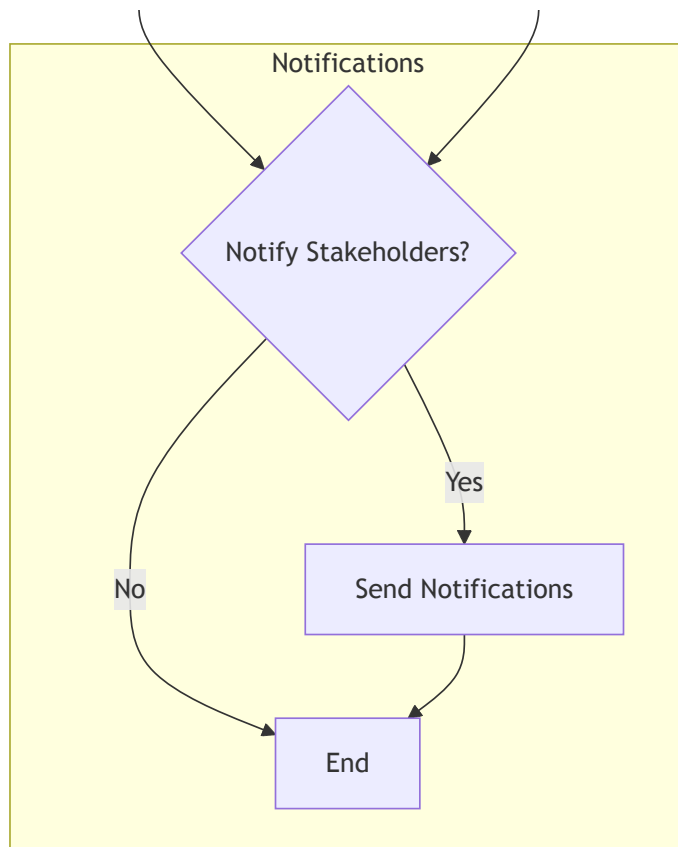




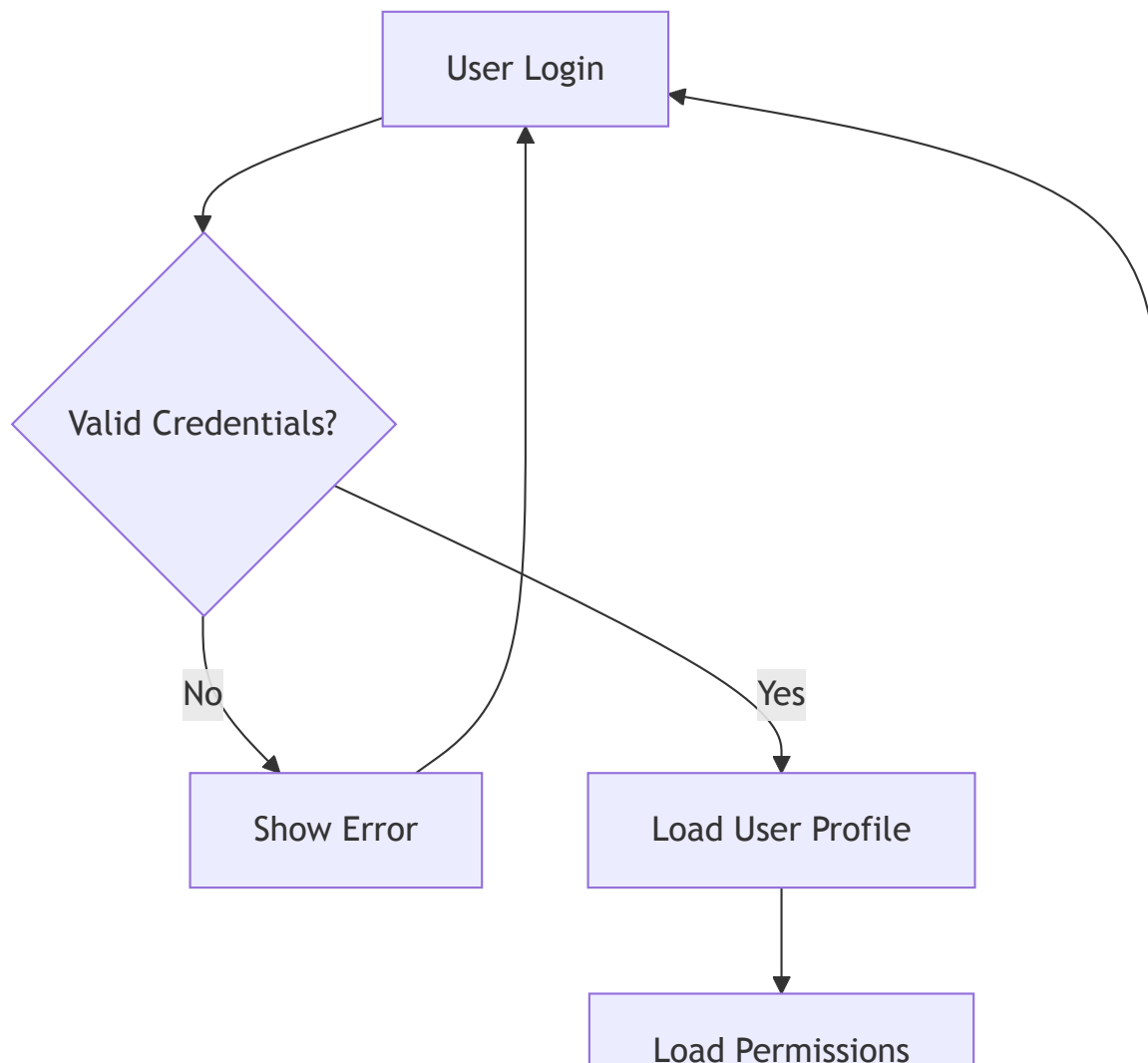
## 4.4 Event/Milestone Tracking Flow

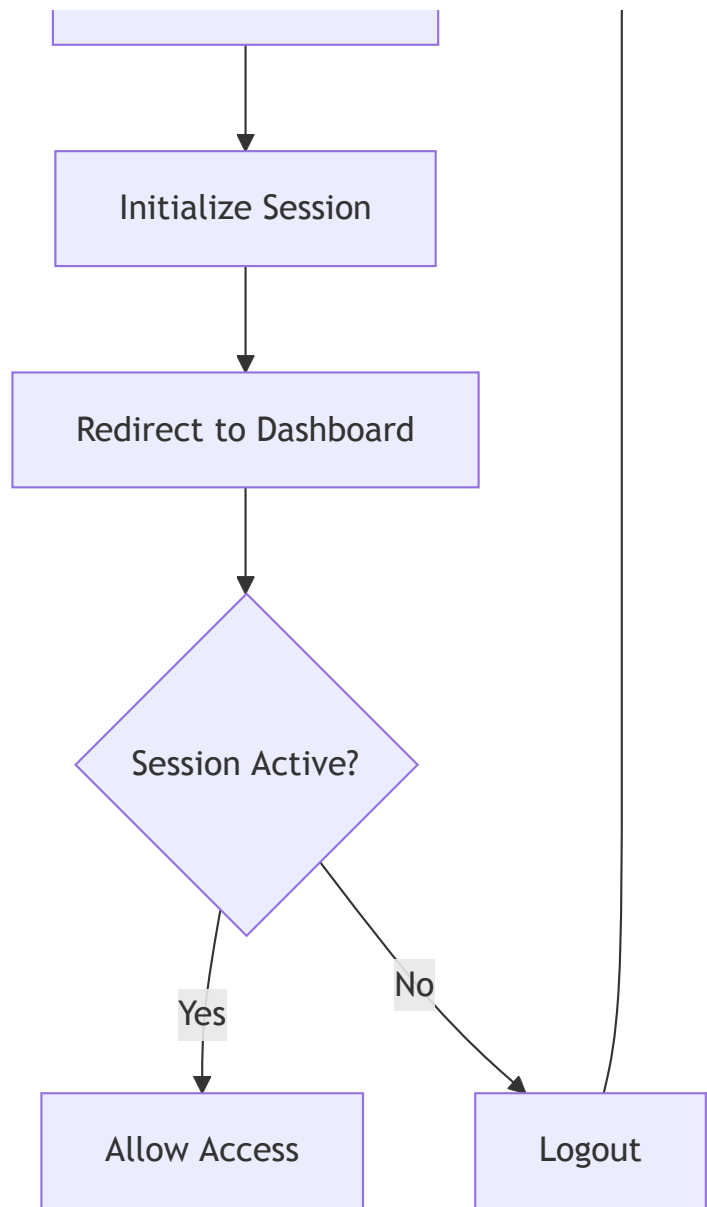






## 4.5 User Authentication Flow

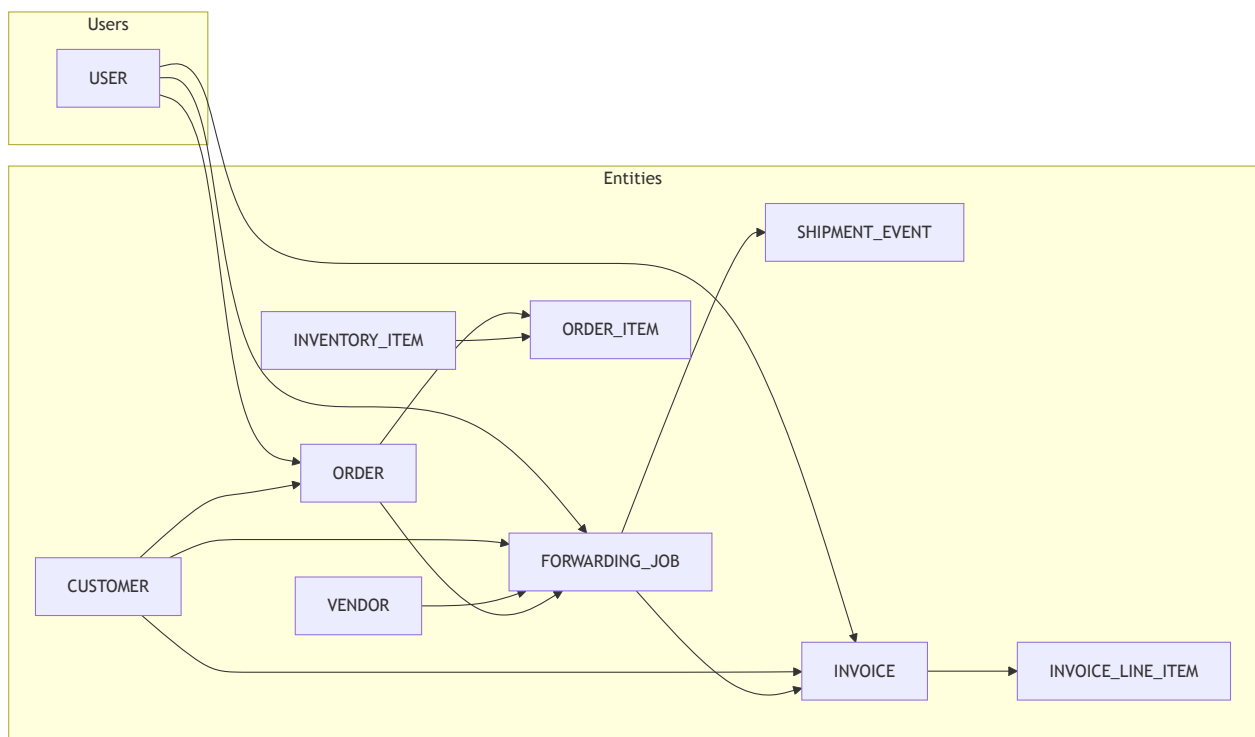




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## 5. Module Relationships

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## 6. UI Components Required

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### 6.1 Layout Components

- **Main Layout:** Header + Left Sidebar Navigation + Main Content Area
- **Header:** App Title, User Profile (Avatar + Name), Notifications Bell, Settings
- **Sidebar Navigation:** Collapsible menu with icons + text labels
- **Content Area:** Dynamic panel with breadcrumbs

### 6.2 Data Display Components

- **Data Grid/Table:**
  - Sortable columns with indicators
  - Row selection (checkboxes)
  - Pagination (items per page, total count)
  - Search bar
  - Filter dropdowns
  - Column visibility toggle
  - Export button
  - Add New button
  - Inline actions (Edit, Delete, View)

## 6.3 Form Components

- **Form Layout:** Multi-column responsive layout
- **Input Fields:** Text, Number, Date, DateTime, Dropdown, Textarea
- **Tab Navigation:** Tabbed forms for complex entities
- **Action Buttons:** Save, Cancel, Delete
- **Validation:** Required fields, format validation

## 6.4 UI Patterns Observed

- **Color Scheme:** Blue header/accents, dark sidebar, light content
- **Typography:** Professional, readable fonts
- **Data Density:** Compact grids for efficiency
- **Responsive:** Desktop-first design

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# 7. Feature Requirements

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### 7.1 Core Features

| Feature             | Priority | Description                                        |
|---------------------|----------|----------------------------------------------------|
| User Authentication | P0       | Login, logout, session management                  |
| Shipment CRUD       | P0       | Create, read, update, delete shipments             |
| Shipment Tracking   | P0       | Real-time status updates, tracking history         |
| Customer Management | P0       | Customer CRUD operations                           |
| Order Processing    | P1       | Order creation and lifecycle management            |
| Vendor Management   | P1       | Carrier/vendor database                            |
| Dashboard           | P1       | KPIs, summary statistics, charts                   |
| Search & Filter     | P0       | Global search, advanced filtering                  |
| Pagination          | P0       | Server-side pagination with configurable page size |
|                     |          |                                                    |

|             |    |                                |
|-------------|----|--------------------------------|
| Data Export | P1 | CSV/Excel export functionality |
|-------------|----|--------------------------------|

## 7.2 Advanced Features

| Feature       | Priority | Description                       |
|---------------|----------|-----------------------------------|
| Invoicing     | P2       | Invoice generation and management |
| Collections   | P2       | Payment tracking and reminders    |
| Inventory     | P2       | Stock management                  |
| Reports       | P1       | Custom report generation          |
| User Roles    | P1       | Role-based access control         |
| Notifications | P2       | In-app and email notifications    |
| Audit Logging | P2       | Track all changes                 |

# 8. Technical Recommendations for Web Implementation

## 8.1 Frontend Stack

- **Framework:** React with TypeScript or Next.js
- **State Management:** React Query for server state, Zustand for UI state
- **UI Library:** Shadcn/ui or Material-UI for components
- **Data Grid:** AG Grid, TanStack Table, or similar for complex tables
- **Forms:** React Hook Form with Zod validation
- **Styling:** Tailwind CSS or CSS Modules

## 8.2 Backend Stack

- **API:** REST API or GraphQL
- **Database:** PostgreSQL for relational data
- **ORM:** Prisma or Drizzle
- **Authentication:** JWT with refresh tokens
- **File Storage:** S3-compatible storage for documents

## 8.3 Key API Endpoints

### # Forwarding Jobs

|        |               |                                          |
|--------|---------------|------------------------------------------|
| GET    | /api/jobs     | - List with pagination, search, filter   |
| POST   | /api/jobs     | - Create new forwarding job              |
| GET    | /api/jobs/:id | - Get job details with parties & routing |
| PUT    | /api/jobs/:id | - Update job                             |
| DELETE | /api/jobs/:id | - Delete job                             |

### # Shipment Events

|        |                      |                         |
|--------|----------------------|-------------------------|
| GET    | /api/jobs/:id/events | - List events for a job |
| POST   | /api/jobs/:id/events | - Add event to job      |
| PUT    | /api/events/:id      | - Update event          |
| DELETE | /api/events/:id      | - Delete event          |

### # Customers

|        |                    |                        |
|--------|--------------------|------------------------|
| GET    | /api/customers     | - List customers       |
| POST   | /api/customers     | - Create customer      |
| GET    | /api/customers/:id | - Get customer details |
| PUT    | /api/customers/:id | - Update customer      |
| DELETE | /api/customers/:id | - Delete customer      |

### # Vendors

|      |                  |                                |
|------|------------------|--------------------------------|
| GET  | /api/vendors     | - List vendors/carriers/agents |
| POST | /api/vendors     | - Create vendor                |
| GET  | /api/vendors/:id | - Get vendor details           |
| PUT  | /api/vendors/:id | - Update vendor                |

### # Orders

|      |                 |                     |
|------|-----------------|---------------------|
| GET  | /api/orders     | - List orders       |
| POST | /api/orders     | - Create order      |
| GET  | /api/orders/:id | - Get order details |
| PUT  | /api/orders/:id | - Update order      |

### # Invoices

|      |                            |                               |
|------|----------------------------|-------------------------------|
| GET  | /api/invoices              | - List invoices with aging    |
| POST | /api/invoices              | - Create invoice              |
| GET  | /api/invoices/:id          | - Get invoice with line items |
| PUT  | /api/invoices/:id          | - Update invoice              |
| POST | /api/invoices/:id/payments | - Record payment              |

### # Reports

|     |                        |                            |
|-----|------------------------|----------------------------|
| GET | /api/reports/aging     | - Aging report by customer |
| GET | /api/reports/shipments | - Shipment summary report  |

### # Auth

|      |                  |                    |
|------|------------------|--------------------|
| POST | /api/auth/login  | - User login       |
| POST | /api/auth/logout | - User logout      |
| GET  | /api/auth/me     | - Get current user |



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## 9. Status Values Reference

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### Forwarding Job Statuses

- Draft, Confirmed, In Progress, At Origin, In Transit, At Destination, Delivered, Closed, Cancelled

### Shipment Event Statuses

- Pending, Completed, Overdue, Skipped, Cancelled

### Order Statuses

- Draft, Pending, Confirmed, Processing, Shipped, Completed, Cancelled

### Invoice Statuses

- Draft, Open, Sent, Partial, Paid, Overdue, Cancelled, Disputed

### Priority Levels

- Low, Medium, High, Urgent

### Standard Milestone Codes

- BKD (Booked), RCV (Received), GIN (Gate In), DOC (Documents), CUS (Customs Submitted)
- CLR (Cleared), LOD (Loaded), DEP (Departed), ARR (Arrived), DIS (Discharged)
- AVL (Available), OFD (Out for Delivery), DLV (Delivered), POD (Proof of Delivery)